

$V_{\text{initial}} = 0\text{V}$ (fully discharged) or whatever starting voltage

I = charging current (realistic average)

Circuit diagram

Fig. 2 shows the circuit diagram of the batteryless microcontroller system. The design is built around the LTC3588, LTC3105, ATtiny13A, SSD1306 OLED display, and supporting components.

Because the ATtiny13A lacks native I²C support, a software-based I²C interface is implemented via bit-banging, enabling reliable communication with the OLED display. The microcontroller is connected to the supercapacitor through a switch. Once the supercapacitor is sufficiently charged, the switch powers both the MCU and the display.

Energy-harvesting sources such as the PV cell and piezo element are connected through diodes to prevent reverse current flow back into the harvesting circuits.

Construction

Assemble the circuit on a breadboard as shown in the author's prototype in Fig. 1. The core of the circuit is an ATtiny13A microcontroller mounted on the breadboard and connected using short jumper wires to minimise losses during batteryless operation.

The harvesting modules, such as the LTC3588 and LTC3105, are connected to external sources, including a piezo element or PV cell. A small electrolytic capacitor or a supercapacitor is placed near the power module and connected with correct

polarity to ensure reliable charge accumulation.

The OLED display (SSD1306) is mounted at the front of the breadboard for clear visibility. The display interfaces with the ATtiny13A through a software (bit-banged) I²C connection implemented using discrete SDA and SCL jumper wires.

A manual switch is inserted between the supercapacitor and the MCU supply rail. This arrangement allows the supercapacitor to charge fully before powering the microcontroller and display, thereby preventing brownout conditions during start-up. Diodes are placed between the harvesting modules to block reverse current from flowing back into the energy sources.

Testing

After connecting the energy-harvesting sources and allowing the supercapacitor to charge, switching on the MCU demonstrates the key outcome of this system: the microcontroller and OLED display operate entirely without a battery. Once powered, the system starts normally and the display updates as expected, confirming that the harvested energy stored in the supercapacitor is sufficient to sustain operation.

Based on estimated calculations and practical observations, the combined current consumption of the MCU and OLED display in continuous active mode is approximately 13mA. The ATtiny13A typically draws about 3mA at 9.6MHz and 2.7-3V, which aligns with datasheet specifications and measured results. The SSD1306 OLED display

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consumes roughly 10mA under normal conditions, assuming moderate text or icon display with 25-50% of the pixels illuminated at a similar supply voltage. Together, these figures result in an average continuous current draw of approximately 13mA.

However, during multimeter measurements, the observed current was only about 7mA, significantly lower than the estimated 13mA. This indicates better-than-expected efficiency under real operating conditions.

Theoretically, with a 10F supercapacitor and a 13mA load, the runtime should be 3 to 4 minutes. In practice, the OLED continued displaying messages for nearly 25 minutes of continuous operation at an average current of 7mA. Table 4 shows the runtime obtained with a 2.7V, 10F supercapacitor.

EFY note. This extended runtime was observed during testing. The precise cause of the improvement beyond the theoretical estimate has not yet been fully established. **EFY**

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